Analysis of Telephone and Data Systems, Inc.

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Abstract

This paper provides a comprehensive financial analysis of Telephone and Data Systems, Inc. (TDS) for the fiscal year 2024. TDS, a parent company to TDS Telecom, UScellular, and Suttle-Straus, navigated a challenging year marked by strategic shifts and investments in fiber expansion. TDS's stock is primarily owned institutionally, with significant insider ownership by the founding Carlson family. Utilizing a PESTEL framework, the paper evaluates TDS's exposure to political, economic, technological, and environmental risks, while also reviewing its financial performance, including profitability concerns, capital structure, and returns. TDS has underperformed relative to the broader market but has shown relative stability compared to peers. With strategic asset sales and reinvestment plans in place, the company seeks to enhance its core telecom operations, including new initiatives in the Mobile Virtual Network Operator (MVNO) market. The paper concludes that despite current profitability challenges, TDS remains committed to growth and maintaining its long-standing dividend payouts.

I. **Analysis of Telephone and Data Systems, Inc.**

This is a financial analysis of Telephone and Data Systems, Inc. for the fiscal year of 2024. NYSE ticker symbol TDS (Yahoo, 2024). TDS is the parent company of TDS Telecommunications LLC. (TDS Telecom/TDS®), UScellular$_{TM}$, and Suttle-Straus (TDS Inc., 2024). As of 2024 TDS no longer owns OneNeck IT Solutions and Data Centers (TDS Inc, 2024). TDS Telecom, headquartered in Madison, Wisconsin, provides TV, phone, and internet service in thirty continental states, seen in figure 1 (TDS Telecom, 2024).

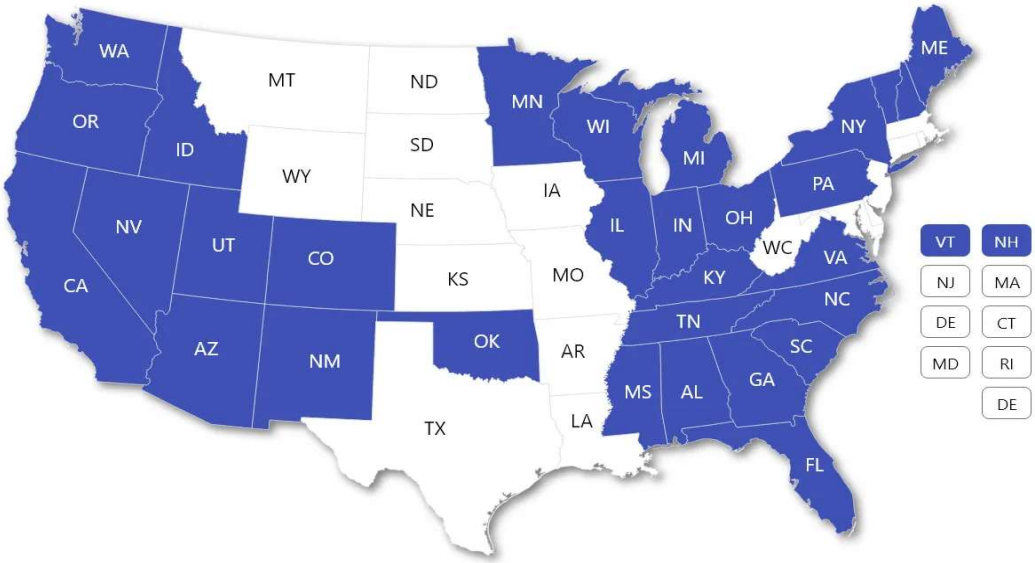

Figure 1: Map of TDS Telecom Operations

UScellular, headquartered in Chicago, Illinois, provides cellular data and voice services nationwide (UScellular, 2024). The map in figure 2 shows coverage which will change with sales of operations to T-Mobile (UScellular, 2024 & T-mobile, 2024). UScellular will retain control of the towers effectively streamlining to a cell placement landlord. Suttle-Straus, headquartered in Waunakee, Wisconsin, provides print materials, mailers, displays and signage, and promotional products (Suttle-Straus, 2024). TDS experienced a good, but turbulent year in 2024.

Figure 2: Map of US Cellular Operations

II. Stockholder Analysis

TDS stockholder breakdown is 80.00% institutionally owned (Marketbeat, 2024). Blackrock Finance, Inc. being the largest investor holding 18.1% (Nasdaq, 2024). Marketbeat also reports insider owned stocks of 14.48% (2024). 96.55% of insider stocks are owned by either the voting trust of the Carlson family or directly or indirectly by members of the Carlson family. Leroy Carlson founded TDS in 1969 (TDS Inc., 2024). This leaves 5.52% for individual retail investors.

Since individual investors are low it is unlikely there are individual marginal investors. When looking at insider marginal investors, individual family members or the voting trust of the Carlson family has considerable influence. When considering institutional marginal investors, Blackrock Finance Inc. is a significant owner. Blackrock Finance is a global investment management firm that offers investment solutions, advisory services, and risk management tools (Blackrock, n.d.)

III. Risk and Return

Using a political, economic, social, technological, environmental, and legal (PESTEL) framework analysis we will identify financial, operational, reputational, market, strategic, environmental, social, governance, and compliance risks, which will be henceforth identified as FORM-SEC risks (Makos, J., 2022). The E borrowed from the well-known ESG acronym. We will ignore the L portion of the framework since currently, TDS faces no reported legal issues other than those associated with the sales of subsidiary companies.

Politically, both TDS Telecom and Uscellular are beholden to changes in "net neutrality" policies of the FCC and fair internet service infrastructure (Hendel, J., 2024). TDS must adhere to FCC rulings as they have committed to the Enhanced Alternative Connect America Fund (E-ACAM) (TDS Telecom, 2023). The E-ACAM program provides additional grants in return for meeting certain requirements in rural areas. Additionally, TDS is affected by changes from standards groups for international telecommunications, like the ITU (Internet Society, 2024).

Economically, S&P Global identified TDS as an intermediate risk, competitively weak in their industry, but positively considered for their capital structure (2024). They also identified TDS as having a significant cash flow to leverage rating and an adequate liquidity (S&P Global, 2024). TDS reported a BB credit rating which indicates some financial stability but makes it prone to economic downturn and changes in the industry (TDS Inc., 2024). Overall CAPEX is down for telecommunications (Dell'oro Group, 2024). TDS announced strategic alternatives in 2023 which led to the sale of one if its subsidiaries in 2024 (TDS Inc., 2024).

TDS is historically known for its small-town beginnings (TDS Inc., 2024). TDS benefits from the needs of customers for local fiber and with satisfaction with streaming services

(American Consumer Satisfaction Index, 2024). TDS' Cares initiative works to give back to communities it is involved with through donations, sponsorships, and volunteerism (TDS Telecom, 2023). This aligns with their growth in new markets and reduces their reputational risks.

TDS has been committed to initiatives aligned with their 2021 ESG report and setup an ESG Committee to follow a "Sustainability Accounting Standards Board (SASB) framework (TDS Inc., 2020)." As a telecommunications company, TDS is dependent on data centers, air-conditioning, and an enormous amount of electric power. This produces a large amount of thermodynamic heat and has a larger overall carbon emission footprint (Monserrate, S. G., 2022).

TDS must navigate a shifting technological landscape with the need of cyber resilience and the rise of generative AI (GAI) (Ernst & Young Global, 2024). Deloitte predicts a GAI growth of 30% (2024). TDS needs to continue improvements to network quality to increase reliability while convincing customers of the benefits of upgrading (Ernst & Young Global, 2024). TDS has recently entered the mobile virtual network operator (MVNO) market (TDS Inc., 2024). But they must also be concerned with low-earth orbit (LEO) and fixed wireless access (FWA) competitors (Deloitte, 2024). Geopolitical conflicts affect supply chains in chip producing countries impacting TDS (Association for Supply Chain Management, 2024). They must also consider the effects of climate change.

TDS has additional FORM-SEC risks listed in the TDS 10-K filing. TDS' smaller scale to larger competitors may cause strain in competition (TDS Inc, 2024). TDS must compete with other telecoms for products, services, pricing, promotions, and network speed (TDS Inc., 2024). Changes in demand for customers can cause churn (TDS Inc, 2024). US Cellular faces changes

in roaming practices that cause revenue to decline and increase roaming expenses (TDS Inc, 2024). US Cellular may struggle with obtaining radio frequencies for future growth (TDS Inc, 2024). Obsolescence of older technologies and deployment of unproven technologies can lead to strain (TDS Inc, 2024). Vendor relationships and financials can also affect TDS (TDS Inc, 2024).

Presuming a purchase of TDS stock for $21.30 at closing for October 2019 and holding to its current closing value of $22.86 the owner would realize a 22.91% total return (Yahoo Finance, 2024). See appendix table 1 for calculations. The stock underperforms in comparison to the Dow Jones Industrial (DJI) market return of 56.97% (Yahoo Finance, 2024). TDS performed at -1.79% for revenue growth over the past 5 years (Yahoo Finance, 2024). When compared to Charter Communications (CHTR), Lumen (LUMN), and Frontier Communications (FYBR), comparable size telecoms, TDS performed better than the average of -11.71%, but significantly lower than CHTR at 19.45% (Stockanalysis.com, 2024). The average indicates that the telecommunications market is experiencing turbulence, but TDS management is reducing the impact. TDS profit margins are down over the last year at -9.31%, indicating they may not be operating efficiently (Stockanalysis.com, 2024). TDS ROE over the past year is -7.24%, much of TDS equity was tied up in ownership of US Cellular (Stockanalysis.com, 2024). Finally, TDS ROA over the past year is 0.92% indicating management is using its assets effectively to generate earnings (Stockanalysis.com, 2024).

When comparing betas of comparable size telecoms with TDS, TDS is less variable with 0.82 over the past 5 years (Stockanalysis.com, 2024). The other three indicate variability higher than one. TDS variability is higher than that of the big three AT&T, Verizon, and T-Mobile (Stockanalysis.com, 2024). See appendix table 2 for calculation. This indicates TDS is managing swings in telecommunications, better than its comparable competitors, but is not as stable as its

largest competitors. The TDS cost of equity is 13.61%, see appendix table 3, indicating the return is higher than the market rate. Mentioned previously, TDS has a BB credit rating indicating it is higher risk. The company has a Debt-to-Equity ratio of 85.86% (Yahoo Finance, 2024). At the end of 2023, TDS shows an interest coverage ratio at -0.95 indicating that it had elevated risk from debt. They also report a 0% Tax rate due to a negative pre-tax income, therefore the cost of debt is equal to the current interest rate (Yahoo Finance, 2024). TDS has a current calculated WACC of 9.14%, seen in appendix table 4, which is higher than the ROIC, seen in appendix table 5, therefore TDS is having difficulty making money for investors.

IV. Measuring Investment Returns

A typical project for TDS is primarily through its subsidiary TDS Telecom. TDS telecom expands its fiber footprint into rural areas. TDS was previously interested in projects of its other subsidiaries from OneNeck holdings or investments in US Cellular (TDS Inc., 2024). TDS reported new fiber-builds in Boise, Idaho, Missoula, Montana and Neenah, Wisconsin in 2024. As an example, when TDS expanded their footprint in Elkhorn, Wisconsin it took rough one and half years reporting launch on May 26, 2021, and completion on January 9, 2023. TDS telecom receives grants from FCC E-ACAM and state Public Service Commissions (PSC). Observing operating and free cash flow in Yahoo Finance you can observe injection of these stimulus funds, with a negative investment cash flow in intangibles alongside and property, plant, and equipment (PPE). A negative free cash period followed this expansion, and then a slowdown in these expenditures. Per financial statements they expand to marketable addresses, but those addresses TDS must convert into paying customers to earn revenue of the expansions. TDS has a new project in flight to enter the Mobile Virtual Network Operator (MVNO) market. Based on telecommunications market recommendations found by both Deloitte and Ernst & Young Global

Limited this may be a solid move. If TDS can complete projects on time and convert fiber addresses to fiber customers, they will increase their revenue. The addition of the MVNO product will create a way for TDS to bundle cellular service with their current bundled internet and TV products.

V. Capital Structure Choices

TDS has several methods at its disposal to raise funds. TDS uses equity financing through both common and preferred stock. They also use employee stock options to compensate employees and reduce the need for cash outflows. This year, TDS entered an agreement with US signal to sell its OneNeck IT and Data center business (TDS Inc., 2024). Also, this year US Cellular sold its wireless operations to T-mobile (T-Mobile, 2024). This has a direct impact on TDS' earnings as TDS is a majority owner. TDS telecom has in the past and could always divest portions of their wireline and broadband networks that are underperforming. According to Buckley TDS is selling off its Texas cable operations as of August this year (Buckley S., 2024).

TDS receives both grants and tax benefits from its involvement in the E-ACAM program through the FCC. Local state PSCs like Wisconsin, through its Broadband Equity, Access, and Deployment (BEAD) program also provide monetary grants for rural expansion. TDS can also receive discounts and vendor financing arrangements like its credit agreement with Nokia (Marketscreener, 2024).

TDS has a Debt-to-equity ratio of 0.9886 which indicates they use both debt and equity financing near equally. Debt shows the market TDS has confidence in its future. Debt payments are predictable. Debt also provides a tax shield. Debt allows TDS a larger amount of capital for investment in its fiber expansion projects. The biggest disadvantage to debt is the requirement to

pay interest payments no matter its market performance. With increased debt also comes an increased credit rating. TDS currently is set at a BB credit rating. If TDS does not bring in enough revenue to cover interest and debt it could risk either default or bankruptcy. Debt puts pressure on management to remain profitable. Market pressures can exacerbate this effect. This pressure can lead to short-term decisions to remain profitable, a reduction in R&D investment, and the postponement of strategic initiatives. Cost-cutting measures could include employee layoffs, hiring freezes, marketing cutbacks, expansion project capex delays, and the sales of assets. Fierce-Networks reported on TDS stating they are "modestly reducing its CAPEX expenditures and they will "shift and pace … construction when necessary" to lessen pressure on their adjusted EBITDA.

VI. Dividend Policy

TDS regularly pays its investors through quarterly dividends. The 2019 quarterly TDS report mentioned a 46-year streak (TDS Inc., 2024). Reviewing 2020 through 2024 the streak continues. No evidence of buybacks was found. TDS earnings have increased from over one million in 1974 to over five billion in 2024 (TDS Inc, 2024 & Yahoo Finance, 2024). Given TDS' history of paying its shareholders, TDS should continue to pay its quarterly dividends and continue its reputation. A slight increase in dividends could reward shareholders. Other funds should be reinvested into TDS' growth strategies.

VII. Valuation

Comparing TDS with telecom competitors of the closest size, Charter, Lumen, Frontier Communications, and Shenadoah Communications, we find that only Charter is above the target value for the past year. See appendix table 1 for numerical comparisons. TDS is below the

median market cap for industry. CHTR is by far the largest company in terms of market cap, SHEN is the smallest. Both TDS and LUMN are trading at below-average market cap levels, indicating potential undervaluation. The EV is more comprehensive as it includes debt, showing that CHTR and LUMN have significantly higher values, consistent with their size. TDS and SHEN remain significantly smaller, with SHEN being far below the group. CHTR stands out with a low EV/EBITDA of 6.85, which might indicate it is undervalued compared to its earnings potential. TDS, LUMN, and SHEN have higher EV/EBITDA ratios, suggesting they are more expensive. TDS and LUMN have lower EV/Sales ratios, indicating potential undervaluation relative to revenue. FYBR and SHEN are valued higher based on sales, with FYBR having the highest ratio, indicating a premium valuation due to growth potential or market sentiment. All firms have exceptionally low P/E ratios, potentially indicating undervaluation or distressed earnings.

TDS has shown a negative growth rate over the last 5 years. TDS has already reported it is looking for strategic alternatives. This negative growth is evidenced by the sales of its OneNeck holdings and the sales of US Cellular's operational division. The US Cellular transaction should close mid-2025, subject to regulatory approval. The OneNeck sale reported completed on September 3rd, 2024 (US Signal, 2024). The proceeds of the same will be seen in Q4. TDS reports it will be utilizing the money from both sales to reinvest in its telecom's fiber growth. The key variable driving TDS Telecom's value is its profit margins.

Summary

This paper was an analysis of Telephone and Data Systems, Inc. (TDS) for 2024. This includes FORM-SEC risks. TDS, which owns TDS Telecom, UScellular, and Suttle-Straus, experienced a turbulent year but continues to invest in fiber network expansion. TDS faces

political, economic, and technological risks, and while its stock has underperformed compared to the broader market, it has shown stability relative to its peers. The company is addressing profitability challenges with strategic sales and reinvestments in its core telecom business. Despite negative profit margins and returns, TDS maintains a strong history of dividend payments and seeks to improve future growth through new ventures like entering the MVNO market and continued rural expansion.

Appendix

Table 1: Comparable Telecoms

	Telephone and Data Systems (TDS)	Charter Communications, Inc. (CHTR)	Lumen Technologies (LUMN)	Frontier Communications Parent (FYBR)	Shenandoah Telecommunications (SHEN)
	Telecom	Telecom	Telecom	Telecom	Telecom
Revenue Start (In Billions)	5.18	45.76	21.46	8.11	0.28738
Revenue End (In Billions)	5.09	54.66	13.72	5.8	0.30862
Revenue Growth (5yr)	-1.74%	19.45%	-36.07%	-28.48%	7.39%
Profit Margins	-9.31%	8.51%	-15.06%	-1.62%	66.75%
Return on Equity (ROE)	-7.24%	34.92%	-150.18%	-1.82%	-1.03%
Return on Assets (ROA)	0.92%	5.36%	0.89%	2.35%	0.18%
Earnings Per Share (EPS)	($4.82)	$31.32	($2.10)	($0.38)	$3.97
Beta (5Y)	0.82	1.05	1.19	1.11	0.62
Comparable Industry and Size					
Taxes	0	0	0	0	0
Depreciation	0.926	8.678	2.997	1.517	0.0775
EBITDA	0.739	20.963	2.08	2.264	0.0791
Market Cap	2.63	46.54	6.3	8.78	0.6968
P/E Ratio	0.5167	0.8514	0.4592	1.5138	2.2578
EV	8.54	143.5	23.69	18.83	0.9965
EV/EBITDA	11.55	6.85	11.39	8.32	13.43
EV/Sales	1.68	2.63	1.73	3.28	3.13

Data From: TDS Inc, 2024, Yahoo Finance, 2024, and Stockanalysis.com, 2024.

Table 2: Large Telecoms, Cellular, Foreign Telecoms, and Tower Companies

	AT&T (T)	**Verizon (VZ)**	**T-Mobile (TMUS)**	**United States Cellular (USM)**	**KT (KT)**	**PLDT (PHI)**	**Crown Castle (CCI)**	**SBA Communications (SBAC)**
	Telecom & Cellular	Telecom & Cellular	Cellular	Cellular	Foreign Telecom	Foreign Telecom	Cell Towers	Cell Towers
Revenue Start (In Billions)	181.19	131.87	45	4.02	24340	169.19	5.76	2.01
Revenue End (In Billions)	122.2	134.24	79.1	3.84	26500	214.5	6.61	2.68
Revenue Growth (5yr)	-32.56%	1.80%	75.78%	-4.48%	8.87%	26.78%	14.76%	33.33%
Profit Margins	10.41%	8.38%	11.95%	1.85%	4.04%	12.39%	18.01%	19.22%
Return on Equity (ROE)	11.79%	12.10%	14.73%	1.72%	5.52%	23.02%	18.74%	n/a
Return on Assets (ROA)	3.96%	4.98%	5.01%	1.08%	2.01%	4.47%	3.65%	7.99%
Earnings Per Share (EPS)	$1.74	$2.66	$7.98	$0.83	$3.15	$2.10	$2.74	$4.75
Beta (5Y)	0.74	0.41	0.51	0.48	0.39	0.3	0.86	0.67

Data From: TDS Inc, 2024, Yahoo Finance, 2024, and Stockanalysis.com, 2024.

Table 3: Cost of Equity

Cost of Equity(re)=rf+β(rm−rf)	
re = Cost of equity	13.61%
rf = Risk-free rate (typically the yield on long-term U.S. Treasury bonds, e.g., 10-year bond)	4.06%
β = Stock's beta	0.82
rm = Expected market return (average market return, often 7%-10%)	15.71%
rm−rf = Equity risk premium (ERP), the excess return expected from investing in the stock market over the risk-free rate.	11.65%
β(rm−rf)	9.55%

Data From: TDS Inc, 2024 and Yahoo Finance, 2024.

Table 4: WACC

Weighted Average Cost of Capital (WACC)	9.14%
Current Stock Price	$23.13
Total Shares Outstanding	106,330,000
E = Market value of equity (total shares outstanding × current stock price)	$ 2,459,412,900.00
D = Market value of debt (total debt from the balance sheet)	$ 5,143,000,000.00
r_e = Cost of equity (from CAPM)	13.61%
r_d = Cost of debt (after tax)	7%
Tax Rate = The company's effective tax rate	0
Income Tax Expense	0
Pre-Tax Income	$ (446,000,000.00)
E+D	$ 7,602,412,900.00
E/(E+D)	0.3235
(E/(E+D))*r_e	0.0440
D/(E+D)	0.6765
(D/(E+D))*r_d	4.74%
(D/(E+D))*r_d*(1-Tax Rate)	4.74%

Table 5: ROIC

Return on Invested Capital (ROIC)	-2.13%
NOPAT = Operating Income (EBIT) × (1 - Tax Rate)	-233,000
Total Debt	5,143,000
Total Equity	6,008,000
Cash	236,000
Invested Capital = Total Debt + Total Equity - Cash (or use Total Assets - Current Liabilities for a simplified version)	10,915,000

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